

Dr. CHINMAY DEHERI

Renewable Energy || Green Hydrogen || Bioenergy Conversion|| Research

Email ID : chinmaydeheri@gmail.com

Mobile No : +91-9668981561

Linkedin ID  www.linkedin.com/in/chinmaydeheri

Google scholar ID  [GmkJNF4AAAAJ](https://scholar.google.com/citations?user=GmkJNF4AAAAJ)

ORCID  <https://orcid.org/0000-0002-5881-3624>

Web of Science ID  [JTU-8658-2023](https://www.webofscience.com/wos/author/uri?uri=JSTN:J1234567890)

SCOPUS Author ID  [56998777200](https://scopus.com/authors/details/scopusid/J1234567890)



Professional Experience: 12+ Years

Associate Professor (Mechanical Engineering) – Vikash Institute of Technology, Bargarh

Duration: December 2023 – Present

Responsibilities:

- Head of the department.
- IQAC and NIRF coordinator.
- Faculty in-charge for R&D and Incubation cell.
- SPOC for Digital Skilling.
- Classroom teaching with project coordination.

Assistant Professor (Mechanical Engineering) – Vikash Institute of Technology, Bargarh

Duration: October 2021 – December 2023

Responsibilities:

- SPOC for Digital Skilling.
- NIRF and IQAC coordinator.
- Startup and Incubation cell coordinator.
- Classroom teaching with project coordination.

Research Scholar- ITER, Siksha 'O' Anusandhan Deemed to be University, Bhubaneswar

Duration: August 2018 – September 2021

Responsibilities:

- Research work on “Biohythane production and conversion for power generation from waste biomass and its statistical evaluation using machine learning”.
- Data Analysis using Machine learning algorithms and Python.
- Research paper writing.
- Project proposal writing.

Assistant Professor (Mechanical Engineering) – Vikash Institute of Technology, Bargarh

Duration: July 2015 – August 2018

Responsibilities:

- Head in the department of Mechanical Engineering.
- Academic coordinator for monitoring the classes.
- Classroom teaching with project coordination.

Lecturer (Mechanical Engineering) – Sushree Institute of Technical Education, Bolangir

Duration: August 2011 – July 2013

Responsibilities:

- Classroom teaching.
- Project coordination to final year students.
- Setting question papers and assignments.

Educational Qualification:

Degree	Institute/University	Passing Year
Ph.D.	ITER, Siksha 'O' Anusandhan Deemed to be University, Bhubaneswar, Odisha	2023
M.Tech	ITER, Siksha 'O' Anusandhan Deemed to be University, Bhubaneswar, Odisha	2015
B.Tech	PKACE, Biju Pattnaik University of Technology, Odisha	2010
+2 Science	CHSE, Odisha	2005
10 th	BSE Odisha	2003

Publications

Journals

1. **Chinmay Deheri**, Saroj Kumar Acharya. Experimental Investigation of Biohythane Performance on Thermal Barrier-Coated Compression Ignition Engine. Journal of Energy Resources Technology. 2023;145(1):011702. **SCI (ASME), I.F. 3.07**
2. **Chinmay Deheri**, Saroj Kumar Acharya. Experimental and statistical evaluation of biohythane fueled thermal barrier coated engine using machine learning algorithms. International Journal of Global Warming. 2023, Vol. 31, Issue 1, pp. 42-67. **SCI (Inderscience). I.F. 1**
3. **Chinmay Deheri**, Saroj Kumar Acharya. Hydrogen generation from anaerobic co-digestion and statistical evaluation using machine learning algorithms. International Journal of Global Warming. 2023, Vol. 30, Issue 2, pp. 152-160. **SCI (Inderscience). I.F. 1**
4. **Chinmay Deheri**, Saroj Kumar Acharya. “Purified biohythane (biohydrogen+biomethane) production from food waste using $\text{CaO}_2 + \text{CaCO}_3$ and NaOH as additives,” International journal of hydrogen energy. 2022, Vol. 47, Issue 5, pp. 2862-2873. **SCI (Elsevier), I.F. 7.139**
5. **Chinmay Deheri**, Saroj Kumar Acharya. “Effect of single and multi-additives on the performance of hydrogen fermentation from food waste,” International Journal of Global Warming. 2022, Vol. 26, Issue 4, pp. 405 – 416. **SCI (Inderscience). I.F. 1**
6. **Chinmay Deheri**, Saroj Kumar Acharya. “Effect of calcium peroxide and sodium hydroxide on hydrogen and methane generation during the co-digestion of food waste and cow dung,” Journal of Cleaner Production. 2021, Vol. 279, pp. 123901. **SCI (Elsevier), I.F. 11.072**

7. **Chinmay Deheri**, Saroj Kumar Acharya, Dharendra Nath Thatoi, Ambica Prasad Mohanty. “A review on performance of biogas and hydrogen on diesel engine in dual fuel mode,” *Fuel*. 2020, Vol. 260, pp. 116337. **SCI (Elsevier), I.F. 8.035**
8. **Chinmay Deheri**, Saroj Kumar Acharya. “An experimental approach to produce hydrogen and methane from food waste using catalyst,” *International journal of hydrogen energy*. 2020, Vol. 45, Issue 35, pp. 17250-17259. **SCI (Elsevier), I.F. 7.139**
9. **C. Deheri**, P.P. Patnaik, S.K. Acharya, D.N. Thatoi. Effect of Ethanol and FeCl₃ Addition on the Performance and Combustion Characteristics of a Compression Ignition Engine. *TEST Engineering and Management*. 2020;83:7575-7584.
10. S.P. Jena, S.K. Acharya & **C. Deheri**. “Thermodynamic analysis of a twin cylinder diesel engine in dual fuel mode with producer gas”. *Biofuels*. 2016, Volume 7, Issue 1, Pages 49-55. **Scopus (Taylor & Francis)**.

Conference Papers

1. **Chinmay Deheri**, Saroj Kumar Acharya, Ruby Mishra. Effect of initial pH and substrate to inoculum ratio on the production of methane and hydrogen from food waste. *Materials Today: Proceedings*. 2023. Volume 74, pp. 650–654. **Scopus (Elsevier)**.
2. **Chinmay Deheri**, Ambica Prasad Mohanty, Saroj Kumar Acharya. “An experimental approach to produce hydrogen from food waste, cow dung, and sludge solution,” *Materials Today: Proceedings*. 2021. Volume 41, pp. 242–246. **Scopus (Elsevier)**.
3. P.P. Patanaik, C.K. Sethi, **C. Deheri**, S.K. Acharya, D.N. Thatoi, Siddharth Som. “An experimental investigation on emission parameters of CI engine using additives,” *Materials Today: Proceedings*. 2021. Volume 41, pp. 280–285. **Scopus (Elsevier)**.

Book Chapters

1. Chinmay Deheri, Binayak Pattanayak, Abinash Mahapatro, Saroj Kumar Acharya. “Grid-Scale Hydrogen Energy Systems Projects and Implementation”, *Hydrogen Energy Production and Fuel Generation*. 2025. pp. 489-508. **Scopus (Wiley Online Library)**.
2. **Chinmay Deheri**, Saroj Kumar Acharya. “Experimental evaluation of biohythane fueled thermal barrier coated dual fuel engine”, *Current Advances in Mechanical Engineering*. 2024. pp. 381-397, **Scopus (Springer Nature)**.
3. Ambica Prasad Mohanty, **Chinmay Deheri**, Saroj Kumar Acharya. “Production of hydrogen from food waste: A review,” *Current Advances in Mechanical Engineering*. 2021. pp. 159-169. **Scopus (Springer Nature)**.

Patent

1. **Chinmay Deheri**, Saroj Kumar Acharya. “Method for Production of Biogas as Renewable Energy Source,” Indian Patent Application No. 202031053416. Published on 10/06/2022.

Awards

1. University Foundation Day Research Award 2023 by Biju Patnaik University of Technology (BPUT), Odisha.

Professional Membership

Life member (LM 143998) of the Indian Society for Technical Education (ISTE) since 2025.

PhD Supervision

Recognized supervisor of the PhD programme as per the BPUT PhD regulation 2023.

Reviewer for various journals

Reviewed more than 10 research papers for various publishers such as Elsevier, Springer Nature, Taylor and Francis, and many more.

Paper Presentations

1. “Experimental evaluation of biohythane fueled thermal barrier coated dual fuel engine”, Recent Advances in Mechanical Engineering Research and Development (ICRAMERD 2023), 20th-22nd July 2023.
2. “Effect of initial pH and substrate to inoculum ratio on methane and hydrogen production from food waste”, Recent Advances in Mechanical Engineering Research and Development (ICRAMERD 2022), 12th-13th August 2022.
3. “Effect of the inoculum ratio to produce methane during the anaerobic digestion of food waste,” Recent Advances in Mechanical Engineering Research and Development (ICRAMERD 2020), 24th-25th July 2020.
4. “An experimental approach to produce hydrogen from food waste, cow dung, and sludge solution,” Recent Advances in Mechanical Engineering Research and Development (ICRAMERD 2020), 24th-25th July 2020.

Professional Training

STTP/FDP

1. Biju Patnaik University of Technology Odisha, 21st-25th April 2025.
Description: “Viksit Bharat: Leadership and Strategic Management (VBLSM)” organized by Biju Patnaik University of Technology, Odisha in association with Association of Indian Universities (AIU).
2. Biju Patnaik University of Technology Odisha, 14th-19th April 2025.
Description: “Empowering Higher Education Institutions in Technology-enabled Learning and Blended Learning” jointly organised by the Commonwealth Educational Media Centre for Asia, New Delhi, the Association of Indian Universities, New Delhi and the Biju Patnaik University of Technology, Odisha.
3. Malaviya Mission - Teacher Training Programme (MM-TTP) of University Grants Commission (UGC), MoE, GoI, 1st-10th March 2025.
Description: “ NEP-2020: Orientation & Sensitization” under Malaviya Mission - Teacher Training Programme (MM-TTP) of University Grants Commission (UGC), MoE, GoI, organized by UGC- MMTTC, JNV University, Jodhpur, Rajasthan.

4. ITER (SOA, Deemed to be University) Bhubaneswar, 15th-22nd September 2020.
Description: “Hybrid and Fuel Cell Vehicle with Advances in Automotive Technology” organized by Department of Mechanical Engineering, ITER (SOA, Deemed to be University), Bhubaneswar in association with Automotive Research Association of India (ARAI), Pune, India.
5. ITER (SOA, Deemed to be University) Bhubaneswar, 23rd-27th September 2020.
Description: “Robotics Design and Application” organized by Department of Mechanical Engineering, ITER (SOA, Deemed to be University), Bhubaneswar in association with Tetrahedron Manufacturing Services, Gurgram, Haryana, India.

International Workshop

1. ITER (SOA, Deemed to be University) Bhubaneswar, 8th-9th January 2021.
Description: “Design, Construction, and Operation of Long Distance Slurry Pipelines for Mineral Transportation” organized by Department of Mechanical Engineering, ITER (SOA, Deemed to be University), Bhubaneswar in association with Atha Group, Kolkata, India.

Internship Courses

1. Pantech Prolabs, India. 1st May-30th May 2021.
Description: “Internship on Machine Learning”
2. Pantech Prolabs, India. 15th May-14th June 2021.
Description: “Internship on Artificial Intelligence”

Skills and Strengths

Enthusiastic and Dedicated work attitude · Educational Management · Relationship Building · Project Planning · Skill Development · Time Management · Leadership Quality · Team Player · Project Management · Analytical Skills · Problem Analysis · Outcome Focused · Impact Assessment · Artificial Intelligence · Machine Learning · Python · Ansys · Matlab · LaTeX · Microsoft Office.

Personal Details :

Date of Birth: 05 July 1988

Marital Status: Married

Languages: English, Hindi, Odia

Interests: Listen to music, Cricket

Permanent address: Sauntamal, Dunguri, Bargarh, Odisha-768052, India